

Model Validation, Overfitting Control & Hyperparameter Tuning

Sumerix Global — AI & ML Internship — Project Task-3

1. Introduction

Task-2 established a baseline House Price Prediction pipeline on the California Housing Dataset using Linear Regression, Ridge Regression, and a Decision Tree, evaluated with a single 80/20 train-test split. Task-3 extends this pipeline to focus on model **reliability** rather than raw accuracy: detecting overfitting, validating performance with 5-fold cross-validation, and systematically tuning the Decision Tree's hyperparameters using GridSearchCV. All models were trained on the same scaled features and the same 80/20 train-test split (random_state=42) to keep comparisons fair.

2. Overfitting Analysis

An unconstrained Decision Tree Regressor was trained first, with no limits on depth or leaf size, to expose how easily tree-based models overfit.

Metric	Untuned Decision Tree
Train RMSE	0.0000
Test RMSE	0.7030
Gap (Test – Train)	0.7030

A Train RMSE of 0.0000 means the tree memorized the training set exactly — every training point ended up in its own leaf. The Test RMSE of 0.7030 is far higher, producing a full 0.7030 gap between the two. This is the textbook signature of overfitting: the model captured noise specific to the training rows instead of the underlying price patterns, so it fails to generalize to unseen data. This confirms the brief's expectation that tree-based models overfit badly without depth/leaf constraints.

3. Cross-Validation for Reliable Evaluation

Rather than trusting a single train-test split, the same untuned tree was evaluated with 5-fold cross-validation across the full scaled dataset:

- Mean CV RMSE: **0.8957** (± 0.0392)

The mean CV RMSE (0.8957) is higher than the single-split Test RMSE (0.7030) seen above. This is expected and is actually the point of cross-validation: the single 80/20 split happened to be a relatively favorable partition for this model, while the 5-fold average reflects performance across five different partitions and is therefore a more honest, less lucky-split-dependent estimate. The small standard deviation (0.0392) shows the model's performance is fairly consistent from fold to fold, so this average can be trusted as a stable baseline before tuning.

4. Hyperparameter Tuning with GridSearchCV

GridSearchCV was used to search over combinations of `max_depth` $\in \{3, 5, 7, 10\}$ and `min_samples_split` $\in \{2, 5, 10\}$, using 5-fold cross-validation scored on negative RMSE.

- Best parameters: **max_depth = 10, min_samples_split = 10**

- Best CV RMSE: **0.6366**

The grid search selected the largest depth and split threshold offered (10 and 10), which constrains the tree just enough to stop it from creating a leaf per training row, while still letting it capture meaningful structure in the housing features. The resulting CV RMSE (0.6366) is noticeably lower than both the untuned tree's CV RMSE (0.8957) and its single-split Test RMSE (0.7030), showing the tuned configuration generalizes better, not just fits better.

5. Evaluating the Optimized Model

The best estimator from GridSearchCV was refit on the training set and evaluated on the held-out test set:

Metric	Untuned Tree	Tuned Tree (max_depth=10, min_samples_split=10)
Train RMSE	0.0000	0.4804
Test RMSE	0.7030	0.6454
R ² Score	0.6228	0.6821
Gap (Test – Train)	0.7030	0.1651

Tuning shrank the train-test gap from 0.7030 down to 0.1651 — roughly a 76% reduction. Notably, the tuned tree's Train RMSE actually rose from 0.0000 to 0.4804 (it fits the training data less perfectly), yet its Test RMSE improved from 0.7030 to 0.6454 and its R² rose from 0.6228 to 0.6821. This is the bias-variance trade-off in action: sacrificing some training accuracy (higher bias) bought a large reduction in variance, which is exactly what a model needs to generalize well in production.

6. Model Comparison Summary

The table below compares all models trained in this task, using the same scaled features and the same train-test split throughout:

Model	Train RMSE	Test RMSE	R ² Score
Linear Regression	0.7197	0.7456	0.5758
Ridge Regression	0.7197	0.7456	0.5758
Untuned Decision Tree	0.0000	0.7030	0.6228
Tuned Decision Tree	0.4804	0.6454	0.6821

Linear Regression and Ridge Regression perform almost identically (Ridge's L2 penalty had negligible effect here, suggesting multicollinearity was not a major issue for the linear model to begin with). Both linear models trail the Tuned Decision Tree on every metric. The Tuned Decision Tree row above is highlighted as the best-performing model overall.

7. Final Model Selection Justification

7.1 Why this model was selected

The **Tuned Decision Tree** (`max_depth=10`, `min_samples_split=10`) is selected as the final model. It has the lowest Test RMSE (0.6454) and the highest R^2 (0.6821) of all four models compared, outperforming Linear Regression and Ridge Regression by roughly 0.10 in RMSE and 10.6 percentage points of R^2 , and outperforming the untuned tree by 0.0576 in RMSE and about 6 percentage points of R^2 — while also being far more reliable, as discussed below.

7.2 How overfitting was reduced

GridSearchCV selected `max_depth=10` and `min_samples_split=10` out of the grids tested (`{3, 5, 7, 10}` and `{2, 5, 10}` respectively). Limiting `max_depth` stops the tree from splitting indefinitely, and requiring at least 10 samples before a further split prevents it from carving out a leaf for a single training row. Together these constraints took the train-test RMSE gap from 0.7030 (untuned) down to 0.1651 (tuned) — about a 76% reduction — directly addressing the overfitting identified in Section 2.

7.3 Why the cross-validation results are trusted

The 5-fold CV RMSE of 0.8957 (± 0.0392) for the untuned tree, and 0.6366 for the best tuned configuration, are both averages over five different train/validation partitions of the data rather than a single lucky or unlucky split. The small standard deviation (0.0392) on the baseline run indicates the model's error is consistent across folds, so the average CV score is a stable estimate of how the model is likely to perform on new, unseen housing data — which is why GridSearchCV was scored on CV RMSE rather than a single held-out score.

7.4 Trade-offs between simplicity and performance

Linear and Ridge Regression are simple, fast, and fully interpretable (each coefficient has a direct meaning), but they underfit the data slightly — their R^2 of 0.5758 shows they only explain about 58% of the variance in house prices, likely because housing price relationships are not purely linear. The untuned Decision Tree can capture non-linear patterns but overfits badly, memorizing noise instead of signal. The Tuned Decision Tree strikes the best balance found in this task: it keeps the flexibility to model non-linear relationships while its depth and split constraints keep it from overfitting, resulting in the best generalization performance of the four models tested. The remaining train-test gap (0.1651) suggests some mild overfitting may still be present, which is the natural trade-off for the extra flexibility a tree offers over a linear model.

8. Conclusion

This task demonstrated why model reliability matters as much as raw accuracy. A single train-test split concealed severe overfitting in the Decision Tree; cross-validation and GridSearchCV together diagnosed and corrected it. The final tuned Decision Tree (`max_depth=10`, `min_samples_split=10`) is recommended as the best-performing and most reliable model from this comparison, with a Test RMSE of 0.6454 and R^2 of 0.6821.